OpsArena: Production Cloud-Native Incident Response RL Environment

TECHNICAL DESIGN DOCUMENT & VERIFIER SPECIFICATION

Team: team_087	**Track:** Software & DevOps / Business Ops	**Evaluated Model:** gpt-6-luna (penpencil.guru)	**Gate:** 100% Passed (8/8 Tasks)

1. Executive Summary & Problem Formulation

Modern cloud operations require autonomous agents to rapidly isolate and mitigate incidents across distributed microservices under tight Service Level Objectives (SLOs: p99 latency < 100ms, error rate < 0.1%), resource constraints, and operational budgets ($150.00 ceiling). Existing benchmarks either present synthetic toy problems or unconstrained Docker sandboxes vulnerable to prompt injection, sandbox escapes, and reward tampering. **OpsArena** provides a deterministic, hardened, discrete-event simulation of **CloudMesh**—an 8-microservice topology—paired with a dual-container evaluation harness, an 8-task deterministic incident suite, and a mathematically gated multi-factor programmatic verifier.

2. Dual-Container Architecture & Sandboxing Threat Model

OpsArena strictly isolates evaluation verifier logic from the execution environment. The harness connects to the external model API and an internal Docker bridge. The sandbox runs an in-memory discrete-event state machine with **zero internet egress**, runs as an unprivileged user (UID 1000), drops ALL capabilities, enforces a 512MB RAM ceiling and 256 PIDs, and never mounts the Docker socket. The agent has **no shell or Python exec access**; all interactions are mediated through 8 typed, schema-validated domain tools. Secret credentials and JWT tokens are scrubbed from logs via automated redaction, and output size is capped at 4,000 characters to eliminate context-stuffing exploits.

3. Operational Action Space (8 Schema-Validated Tools)

Tool	Cost	Parameters	Function & Safeguards
get_topology	$1.00	None	Returns microservice tiers, pod replicas, versions, and active budget spend.
query_metrics	$2.00	service, metric	Queries p99 latency, error rate, CPU, memory, or QPS timeseries history.
query_logs	$3.00	service, filter	Inspects structured application log tails with automated secret sanitization.
scale_service	$2/pod+$3	service, replicas	Scales service replicas between 1 and 20 pods within platform capacity limits.
rollback_release	$20.00	service, version	Rolls back a faulty deployment to a known stable image in the release registry.
update_config	$5.00	service, key, value	Updates runtime config keys (db_pool_size, circuit_breaker_enabled, etc.).
reroute_traffic	$10.00	service, weight	Shifts ingress/canary traffic weights (0% to 100%) to safely isolate bad routes.
finish	$0.00	None	Signals mitigation completion and triggers programmatic verification scoring.

4. Multi-Factor Programmatic Reward Formulation

OpsArena computes a multi-factor continuous reward gated by an absolute incident resolution indicator: **Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]**. Here **P** ∈ **{0, 1}** is a binary predicate indicating whether the primary incident root cause was remediated. If unmitigated, **P = 0**, guaranteeing that no-op or unverified runs earn exactly **0.0000** (strictly ≤ 0.05). **S_SLO** scores latency and error restoration; **R_reg** penalizes crashes on non-target services; **B_bud** rewards operating under $50 (decaying to 0 at $150); and **E_eff** rewards resolving within ≤ 8 actions.

5. Deterministic 8-Task Incident Benchmark Suite

The benchmark features 8 seeded incident scenarios spanning Easy, Medium, and Hard operational challenges:

Task ID	Incident Title	Diff	Root Cause & Simulated Failure	Target Mitigation	Oracle	No-Op
task_01	Redis Eviction Storm	Easy	Cache memory storm causes thundering herd on DB (CPU 96%).	Scale cache-cluster ≥ 2 pods	0.991	0.000
task_02	Bad Deployment Rollback	Easy	Faulty v2.4.1 causes NullPointerExceptions (18.4% 500 errors).	Rollback order-service to v2.4.0	0.996	0.000
task_03	DB Pool Starvation	Med	Checkout bursts exhaust connection pool (504 gateway timeouts).	Update db_pool_size ≥ 25	0.993	0.000
task_04	Slow Third-Party Gateway	Med	Hanging external payment calls cause thread starvation.	Enable circuit_breaker_enabled	0.994	0.000
task_05	Cascading Failure	Hard	Flash traffic surge (4.8k RPS) causes pod OOM crash loop.	Scale order-service ≥ 6 pods	0.983	0.000
task_06	Canary Memory Leak	Med	Memory leak in 10% canary traffic route degrades orders.	Reroute traffic weight to 0%	0.994	0.000
task_07	Overprovisioned Drain	Hard	Platform overscaled (36 pods) with low CPU, burning $450/hr.	Right-size cluster to ≤ 18 pods	0.996	0.000
task_08	Migration Schema Lock	Hard	Exclusive table lock blocks writes (22% checkout errors).	Set lock_timeout_ms=1000	0.991	0.000

6. Empirical Evaluation & Live Autonomous Model Execution

Pre-Flight Baseline Gating: All 8 tasks underwent pre-flight verification against the hackathon gate. Across all tasks, the reference Oracle scored between **0.983 and 0.996** (≥ 0.95 PASS), while the No-Op agent scored exactly **0.000** (≤ 0.05 PASS). Total pre-flight check time was **3.28 seconds**, well within the 120s budget.

Live gpt-6-luna Autonomous Execution: The model was evaluated on the designated demo task (task_02_bad_deployment) via the hackathon endpoint (https://litellm-hackathon.penpencil.guru). The model autonomously inspected system topology and error metrics in Step 1, executed rollback_release in Step 2, re-verified system restoration in Step 3, and called finish in Step 4. Total wall-clock time was **13.7 seconds** with **0 errors**, achieving a final reward of **0.9955 / 1.0000**.

7. Run Contract & Submission Compliance Checklist

Requirement	Specification	OpsArena Implementation Status
Single-Command Execution	docker compose up --build ...	VERIFIED: docker-compose.yml runs sandbox & harness with exit code forwarding.
Baseline Gating Contract	Oracle ≥ 0.95, No-Op ≤ 0.05	VERIFIED: Exit code 2 if baseline fails; 8/8 tasks verified in gate.json (3.28s).
Model Egress & API	OpenAI-compatible via env vars	VERIFIED: Connects to gpt-6-luna via BASE_URL, API_KEY, MODEL in .env.
Resource & Step Limits	40 steps, 10 min, 10 tools/turn	VERIFIED: Strict per-step enforcement, 48K token trimming, 4k char clipping.
Submission Deliverables	Logs, Trajectories, PDF Doc, Deck	COMPLETE: team_087.zip packaged with all required directories and reports.